

Electrostatic Potential and Capacitance

SHORT NOTES

Electric Potential Energy

- (a) The electric potential energy for a point charge q_0 in the electric field of a stationary point charge q , with a distance r separating the charges is

$$U = \frac{1}{4\pi\epsilon_0} \frac{q_0 q}{r}$$

If electric potential energy at infinity is considered to be zero.

- (b) Work done by electric force on a charge when it is moved from A to B

$$W_{A \rightarrow B} = U_A - U_B$$

Electric Potential

- (a) Potential is equal to potential energy per unit charge

$$V = \frac{U}{q_0}$$

The potential for a point charge q at any point at a distance r is

$$V = \frac{1}{4\pi\epsilon_0} \frac{q}{r}$$

If potential at infinity is considered to be zero.

- (b) Potential due to a collection of charges is the sum of the potentials due to each charge.

$$V_n = \frac{1}{4\pi\epsilon_0} \sum_i \frac{q_i}{r}$$

- (c) Potential due to a conducting sphere of radius r with charge q (solid or hollow) at a distance r from the centre

$$V = \left(\frac{1}{4\pi\epsilon_0} \right) \frac{q}{r} \quad \text{if } (r > R)$$

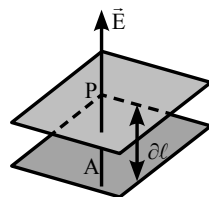
$$V = \left(\frac{1}{4\pi\epsilon_0} \right) \frac{q}{R} \quad \text{if } (r = R)$$

$$V = \frac{1}{4\pi\epsilon_0} \frac{q}{R} \quad \text{if } (r < R)$$

- (d) Relation between electric field and potential

$$|\vec{E}| = \left| -\frac{\partial V}{\partial \ell} \right|$$

$$= + \frac{|\partial V|}{\partial \ell}$$



Electric Dipole Potential

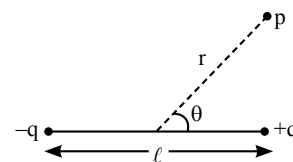
- (a) At axis $V = \frac{1}{4\pi\epsilon_0} \frac{p}{r^2}$

- (b) Equatorial line $V = 0$

$$(c) V = \frac{1}{4\pi\epsilon_0} \left(\frac{p \cos \theta}{r^2} \right)$$

- (d) Potential energy of a dipole in an external electric field

$$U(\theta) = -\vec{p} \cdot \vec{E}$$



Capacitors

Capacitance of a parallel plate capacitor $C = \epsilon_0 \frac{A}{d}$

$$\text{Also, } C = \frac{Q}{V}$$

Electric Field Energy

$$(a) U = \frac{1}{2} QV = \frac{Q^2}{2C} = \frac{1}{2} CV^2$$

- (b) Energy density of energy stored in electric field $U = \frac{1}{2} \epsilon_0 E^2$

Combination of Capacitor

- (a) When capacitors are combined in series,

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \dots$$

- (b) When capacitors are connected in parallel,

$$C_{eq} = C_1 + C_2 + C_3 + \dots$$

- (c) Parallel plate capacitor $C = \frac{A \epsilon_0}{d}$

- (d) If a dielectric medium of dielectric constant K is filled completely between the plates then capacitance increases

$$\text{by } K \text{ times i.e., } C' = \frac{K \epsilon_0 A}{d} \Rightarrow C' = KC$$

- (e) Spherical capacitor,

$$C = 4\pi\epsilon_0 \frac{ab}{b-a}, \quad (\text{When outer shell is earthed})$$

$$C = 4\pi\epsilon_0 \frac{b^2}{b-a}, \quad (\text{When inner shell is earthed})$$

$$C = 4\pi\epsilon_0 R \quad (\text{For a sphere of radius } R)$$

- (f) Cylindrical capacitor, $C = \frac{2\pi\epsilon_0 l}{\ln\left(\frac{b}{a}\right)}$

Dielectrics

- (a) Induced charge, $q' = q \left(1 - \frac{1}{K} \right)$

- (b) Polarisation $P = \frac{\text{Dipole moment}}{\text{Volume}} \Rightarrow \vec{P} = \chi_0 \vec{E} \Rightarrow \frac{\chi_0}{\epsilon_0} = K - 1$